🏆 Infosys SE Elite Crack Vault — 09 Mensuration 2D 3D and Coordinate Geometry Elite

🎯 Infosys Systems Engineer (SE) | 📊 10 High-Yield Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

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Problem 01 • Cube Root & Mensuration Shortcuts

🔥 MUST-SOLVE

`ID: SEC02_BOOK_P05_Q30` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> Find the cube root of 328,509:
> $$\sqrt[3]{328509}$$

- a. 69
- b. 50
- c. 97
- d. 78

✓ **Verified Answer:** `A. 69`

Problem 02 • Cube Root & Mensuration Shortcuts

🔥 MUST-SOLVE

`ID: SEC02_BOOK_P05_Q31` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> Find the cube root of 50,653:
> $$\sqrt[3]{50653}$$

- a. 22
- b. 37
- c. 97
- d. 69

✓ **Verified Answer:** `B. 37`

Problem 03 • Angular Geometry & Directional Rotation

MUST-SOLVE

`ID: SEC02_BOOK_P26_Q199` `Source: 250 + Video Solution Quant Assorted Book.pdf` `Level: Infosys SE Standard`

> Shiva is facing North. He turns 180° and 45° in the clockwise direction, then 270° in the anti-clockwise direction, and finally turns 180°. What is his facing direction now?

- a. SE (South-East)
- b. SW (South-West)
- c. South
- d. NE (North-East)

✓ **Verified Answer:** `A. SE (South-East)`

Problem 04 • 2D Perimeter & Wire Re-bending Invariant

MUST-SOLVE

`ID: SEC02_MENSUR_WIRE_Q01` `Source: CampusMonk Quantitative Aptitude` `Level: Infosys SE Standard`

> A copper wire is bent in the form of a circle of radius 28 cm. If the same wire is straightened and re-bent into the shape of a square, what is the length of each side of the square? (Take $\pi = 22/7$)

- a. 40 cm
- b. 44 cm
- c. 48 cm
- d. 52 cm

✓ **Verified Answer:** `B. 44 cm`

Problem 05 • 3D Cuboid Surface Area & Ratio Scaling

MUST-SOLVE

`ID: SEC02_TEST_MENSUR_Q01` `Source: Mensuration Pre-Test_QUESTIONS.json` `Level: Infosys SE Standard`

> The ratio of the length, width, and height of a closed rectangular cuboid is $7 : 5 : 4$. If its entire surface area (Total Surface Area) is $2,656\text{ cm}^2$, find its width.

- a. 16 cm
- b. 20 cm
- c. 24 cm
- d. 28 cm

✓ **Verified Answer:** `B. 20 cm`

Problem 06 • Cylinder Total Surface Area

CORE SE

ID: SEC02_TEST_MENSUR_Q02 | Source: Mensuration Pre-Test_QUESTIONS.json | Level: Infosys SE Standard

> Find the total surface area (in cm^2) of a right circular cylinder having a diameter of 7 cm and a height of 6 cm. (Take $\pi = 22/7$)

- a. 185 cm^2
- b. 198 cm^2
- c. 209 cm^2
- d. 220 cm^2

✓ **Verified Answer:** `C. 209 cm²`

Problem 07 • 3D Sphere Melting & Recasting Conservation

CORE SE

ID: SEC02_MENSUR_RECAST_Q01 | Source: CampusMonk Quantitative Aptitude | Level: Infosys SE Standard

> A solid metallic sphere of radius 6 cm is melted and recast into small spherical lead balls, each of radius 2 cm. How many such small balls can be made from the parent sphere?

- a. 18
- b. 24
- c. 27
- d. 36

✓ **Verified Answer:** `C. 27`

Problem 08 • Cone and Cylinder Equal Base Volume Comparison

CORE SE

ID: SEC02_MENSUR_CONE_CYL_Q01 | Source: CampusMonk Quantitative Aptitude | Level: Infosys SE Standard

> A right circular cone and a cylinder have equal circular bases and identical heights. If the volume of the cylinder is 180 cm^3, what is the volume of the cone?

- a. 45 cm^3
- b. 60 cm^3
- c. 90 cm^3
- d. 120 cm^3

✓ **Verified Answer:** `B. 60 cm³`

Problem 09 • Circular Track Concentric Pathway Area

CORE SE

ID: SEC02_MENSUR_TRACK_Q01 Source: CampusMonk Quantitative Aptitude Level: Infosys SE Standard

> A circular garden of radius 21 m is surrounded by a running track of uniform width 3.5 m. What is the area of the running track? (Take $\pi = 22/7$)

- a. 462.5 m^2
- b. 485.0 m^2
- c. 500.5 m^2
- d. 525.0 m^2

✓ **Verified Answer:** `C. 500.5 m²`

Problem 10 • Trapezium Field Geometry

CORE SE

ID: SEC02_MENSUR_TRAP_Q01 Source: CampusMonk Quantitative Aptitude Level: Infosys SE Standard

> The parallel sides of an isosceles trapezium are 20 cm and 10 cm. Each of the non-parallel oblique sides is 13 cm. What is the total area of the trapezium?

- a. 150 cm^2
- b. 180 cm^2
- c. 210 cm^2
- d. 240 cm^2

✓ **Verified Answer:** `B. 180 cm²`